

cum-v15

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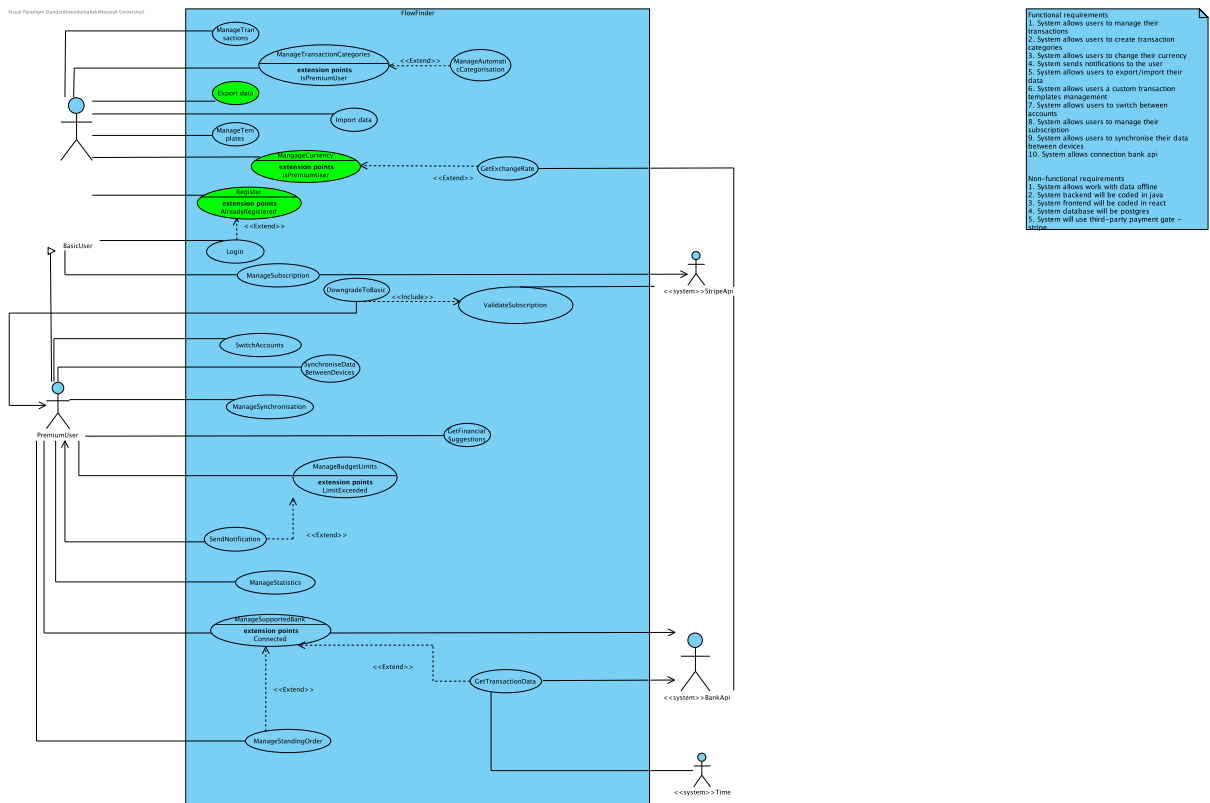
Stereotypes

Case Story		
Context		
requirement		
Tagged Value Definitions	Text	
	Backlog Activity Id	0
	Type	HTML
	ID	
	Backlog Activity Id	0
	Type	Text
	source	
	Backlog Activity Id	0
	Type	Text
	kind	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Functional, Performance, Interface
	verifyMethod	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Analysis, Demonstration, Inspection, Test
	risk	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	High, Medium, Low
	status	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Proposed, Approved, Rejected, Deferred, Implemented, Mandatory, Obsolete
UseCase		
Tagged Value Definitions	Level	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Summary, User, Subfunction

	Complexity	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Low, Medium, High
	Use Case Status	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Name Only, Initial, Base, Complete, Deferred
	Implementation Status	
	Backlog Activity Id	0
	Type	Enumeration
	Enumeration Values	Scheduled, Started, Partially Complete, Complete, Partially Deferred
	Preconditions	
	Backlog Activity Id	0
	Type	Multi-line Text
	Post-conditions	
	Backlog Activity Id	0
	Type	Multi-line Text
	Author	
	Backlog Activity Id	0
	Type	Text
Assumptions		
Backlog Activity Id	0	
Type	Multi-line Text	
Extend		
Include		
Interface		

Use Case Diagram

use-case-diagram



Details

ManageBudgetLimits

Extension Points

LimitExceeded	
Backlog Activity Id	0

ManageSupportedBank

Extension Points

Connected	
Backlog Activity Id	0

Backlog Activity Id	0

Register

Extension Points

AlreadyRegistered	
Backlog Activity Id	0

● ManageCurrency

Extension Points

IsPremiumUser	
Backlog Activity Id	0

● ManageTransactionCategories

Extension Points

IsPremiumUser	
Backlog Activity Id	0

Children

Name	Description
● DowngradeToBasic	Brief description: UC34 downgrades the PremiumUser to BasicUser.
● Import data	Brief description: UC32 can import transactions into the system.
● ManageStatistics	Brief description: UC31 enables the User to see his financial statistics in the form of a graph. Statistics allow for recalculation based on selected currency and exchange rate. User can select to export or print the displayed statistics
● GetFinancialSuggestions	Brief description: UC30 enables the User to select the option to optimize expenses. Application analyses user's data and display financial suggestions.
● ManageAutomaticCategorisation	Brief description: UC22 can set up automatic categorization rules (e.g., "if the transaction name contains 'Albert,' assign it to 'Groceries'") and/or enable an AI model that learns from transaction history to suggest the most likely category automatically.
● SendNotification	Brief description: UC21 will send a notification warning the User that they have exceeded the set budget.
● GetExchangeRate	Brief description: UC20 allows the user to retrieve exchange rates automatically from Bank API
● ManageBudgetLimits	Brief description: UC19 enables the User to set a budget limit for selected categories (e.g., "max. 5000 CZK per month for food"). The application will monitor their compliance, alerting the user if they are exceeded
● ManageStandingOrder	Brief description: UC18 enables the user to create, modify, or cancel standing payment orders within the FlowFinder application. It allows users to automate

	regular payments, such as rent or subscriptions, by setting the amount, frequency, and payment date. The system ensures that payments are executed according to the defined schedule and notifies the user of any issues or upcoming transactions												
● ManageSupportedBank	Brief description: UC17 allows the system administrator to manage the list of banks supported by the FlowFinder application. The User can add, update, or remove bank connections and verify API compatibility. The updated list of supported banks is then made available to users when linking their bank accounts within the app												
● ManageSynchronisation	Brief description: UC14 provides the user with control over synchronization settings, including enabling/disabling auto-sync, selecting data categories to sync, and resolving synchronization conflicts..												
● ValidateSubscription	Brief description: UC11 checks the user's current subscription status to verify its validity, expiration date, and eligibility for accessing premium features.												
● ManageSubscription	Brief description: UC10 enables the user to view, upgrade, downgrade, renew, or cancel their current subscription plan according to available options.												
● Login	Brief description: UC09 allows a registered user to securely access the system by entering valid authentication credentials												
● Register	<table border="1"> <tr> <td>Use Case ID</td><td>8</td></tr> <tr> <td>Brief Description</td><td>UC 08 allows user to register and create an account</td></tr> <tr> <td>Primary Actors</td><td>BasicUser</td></tr> <tr> <td>Secondary Actors</td><td>-</td></tr> <tr> <td>Preconditions</td><td>User is not logged into the system, System is online and available, User does not already have an active account (or at least not with the same email/username)</td></tr> <tr> <td>Main Flow of Events</td><td>1.The user navigates to the registration page 2.The system displays registration form 3.The user enters the required information. 4.The user submits the registration form.</td></tr> </table>	Use Case ID	8	Brief Description	UC 08 allows user to register and create an account	Primary Actors	BasicUser	Secondary Actors	-	Preconditions	User is not logged into the system, System is online and available, User does not already have an active account (or at least not with the same email/username)	Main Flow of Events	1.The user navigates to the registration page 2.The system displays registration form 3.The user enters the required information. 4.The user submits the registration form.
Use Case ID	8												
Brief Description	UC 08 allows user to register and create an account												
Primary Actors	BasicUser												
Secondary Actors	-												
Preconditions	User is not logged into the system, System is online and available, User does not already have an active account (or at least not with the same email/username)												
Main Flow of Events	1.The user navigates to the registration page 2.The system displays registration form 3.The user enters the required information. 4.The user submits the registration form.												

	5.The system validates the input 6.The system checks whether the email/username is already in use. EXTENSION POINT(AlreadyRegistered) 7.If validation is successful 7.1 The system creates a new user account in the database. 8.The system informs the user that the registration was successful.
Alternative Flows	A1: Invalid Input At Step 5, if validation fails, the system displays an error message and prompts the user to correct the data. A2: Duplicate Account At Step 6, if the email/username already exists, the system displays an error message and suggests the user to log in or reset their password. A3: System Error If the system cannot save the BasicUser's data, it displays an error message and advises the user to try again later. The BasicUser may leave the page anytime by clicking on the "Back" button
Post-conditions	Success: A new user account is created and stored in the system; the user is informed of successful registration Failure: No account is created, and the system remains unchanged User cancellation: The process ends with no changes to the system
● GetTransactionData	Brief description: UC09 will periodically download transactions from the supported bank using bank's API.
● SynchroniseDataBetweenDevices	Brief description: UC13 ensures that the user's data is consistent and up-to-date across multiple devices linked to the same account.
● SwitchAccounts	Brief description: UC12 allows the user to switch between multiple accounts without needing to log out completely.
● ManageTemplates	Brief description: UC06 can create and manage transaction templates to simplify recurring entries (e.g., "Salary," "Spotify Subscription," "Grocery

 Export data	Shopping"). When creating a new transaction, a template can be used to auto-fill common fields.	
	Use Case ID	UC05
	Brief Description	UC05 allows to the BasicUser to export transactions to the file.
	Primary Actors	BasicUser
	Secondary Actors	
	Preconditions	System is running and user has access permissions to perform data import/export.
	Main Flow of Events	1.User navigates to the Export section. 2. IF the user selects Export Data: 2.1 System displays export format options. 2.2 User selects the desired export format 2.3 System displays data scope options. 2.4 User selects data scope. 2.5 User confirms the export. 2.6 System generates the export file. 2.7 System stores the file locally. 2.8 System notifies the user that the export has completed successfully.
	Alternative Flows	1: Export Cancellation: User cancels the export process before completion and system aborts the operation and returns to the previous screen without saving a file.
 MangageCurrency	Post-conditions	A valid export file is created and stored in the user-specified location. System logs the export activity for auditing purposes.

Use Case ID	3
Brief Description	UC03 allows the user to define, edit, and remove currencies in the FlowFinder application, as well as to update and apply exchange rates used for transaction conversions.
Primary Actors	BasicUser
Secondary Actors	-
Preconditions	The system is running and the user is able to access the currency management menu.
Main Flow of Events	1. UC is invoked once the User selects Manage Currencies from the application menu 2. FOR EACH (existing currency) 2.1. System displays currency name and its exchange rate 3. The User chooses one of the following actions 4. IF the User chooses to add new currency 4.1. The system prompts the User to enter a currency code, currency name and exchange rate 4.2. The User enters the currency code, currency name and exchange rate EXTENSION POINT(isPremiumUser) 5. IF the User chooses to edit an existing currency 5.1. The system prompts the User to change currency code, currency name, currency exchange rate 5.2. The User updates the currency code, currency name or exchange rate as needed 6. IF the User chooses to remove an existing currency 6.1. The system prompts the User if he really wants to remove the currency

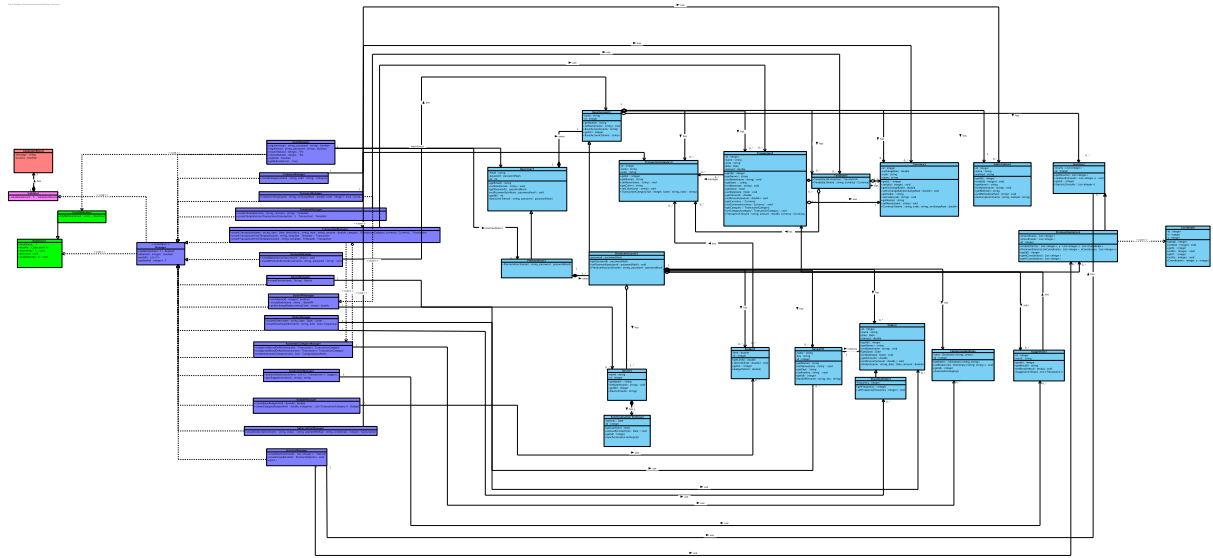
		6.2. The User confirms or cancels the removal action 7. The system validates the input 8. The system saves the changes to the database 9. The system displays the updated list of currencies in the currency management menu
	Alternative Flows	A1. The user cancels the action No changes are applied A2. Input validation fails (e.g., invalid currency code) The system displays an error message and prompts for correction The BasicUser can leave the page anytime by selecting "back" button.
	Post-conditions	List of currencies is updated
● ManageTransactionCategories	Brief description: UC02 can create, edit, and delete transaction categories (e.g., "Rent," "Food," "Salary"). Categories help organize financial data and enable clearer statistical insights.	
● ManageTransactions	Brief description: UC01 can add, edit, and delete financial transactions (income and expenses). Each transaction includes an amount, date, description, currency, note, and category. Transactions can be filtered and displayed in an overview with basic statistics (e.g., total income/expenses).	

Class Diagram



Class Diagram

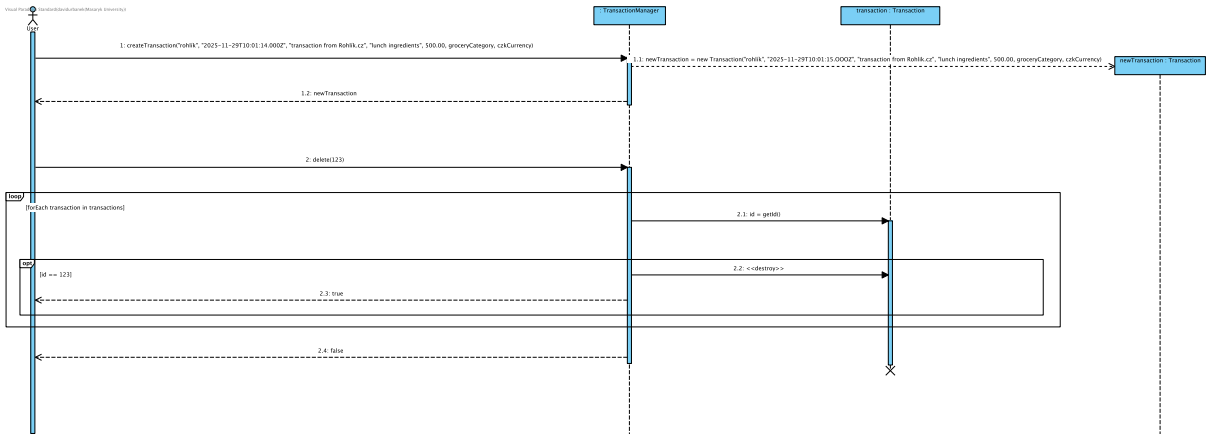
design class diagram



Details

Sequence Diagram

Sequence Diagram1



Details

Children

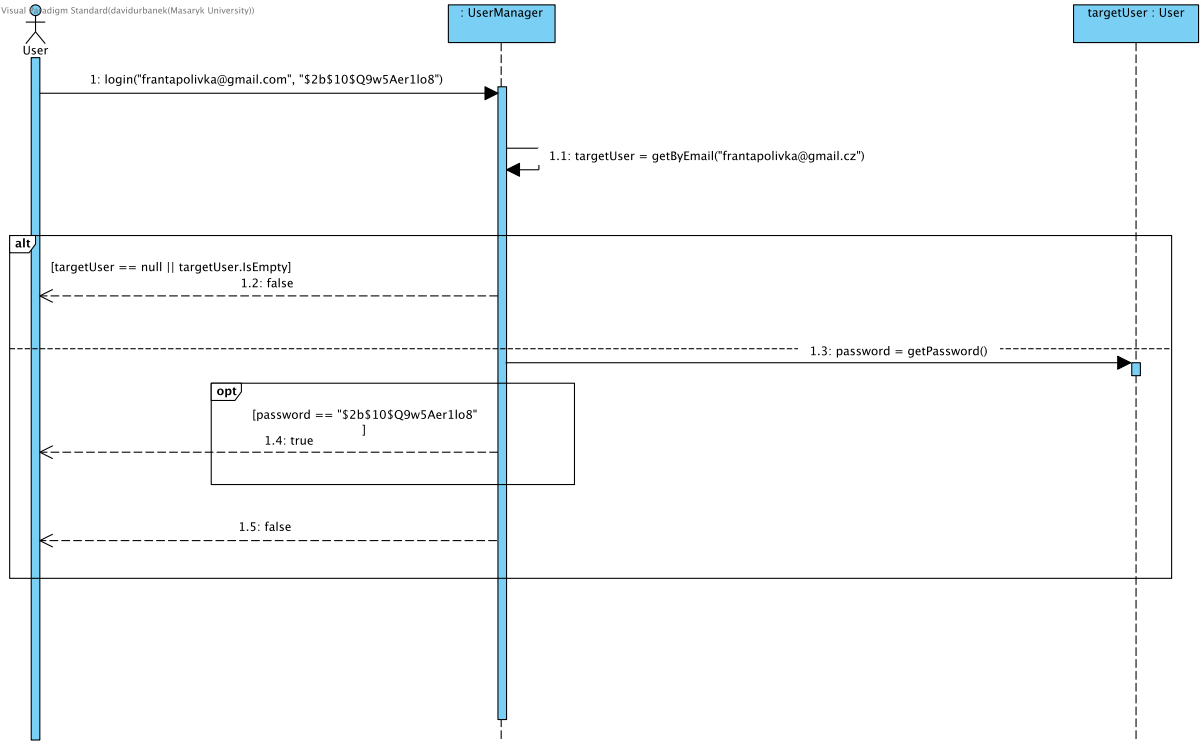
Name	Description
 id == 123	

Sequence Diagram2

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Sequence Diagram

Sequence Diagram3



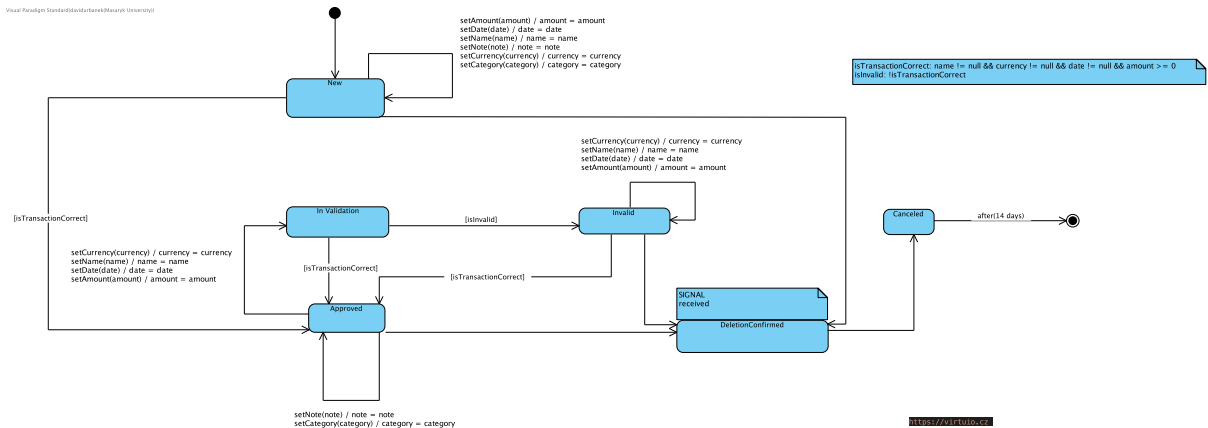
Details

Children

Name	Description
 CombinedFragment2	

State Machine Diagram

State Machine Diagram1



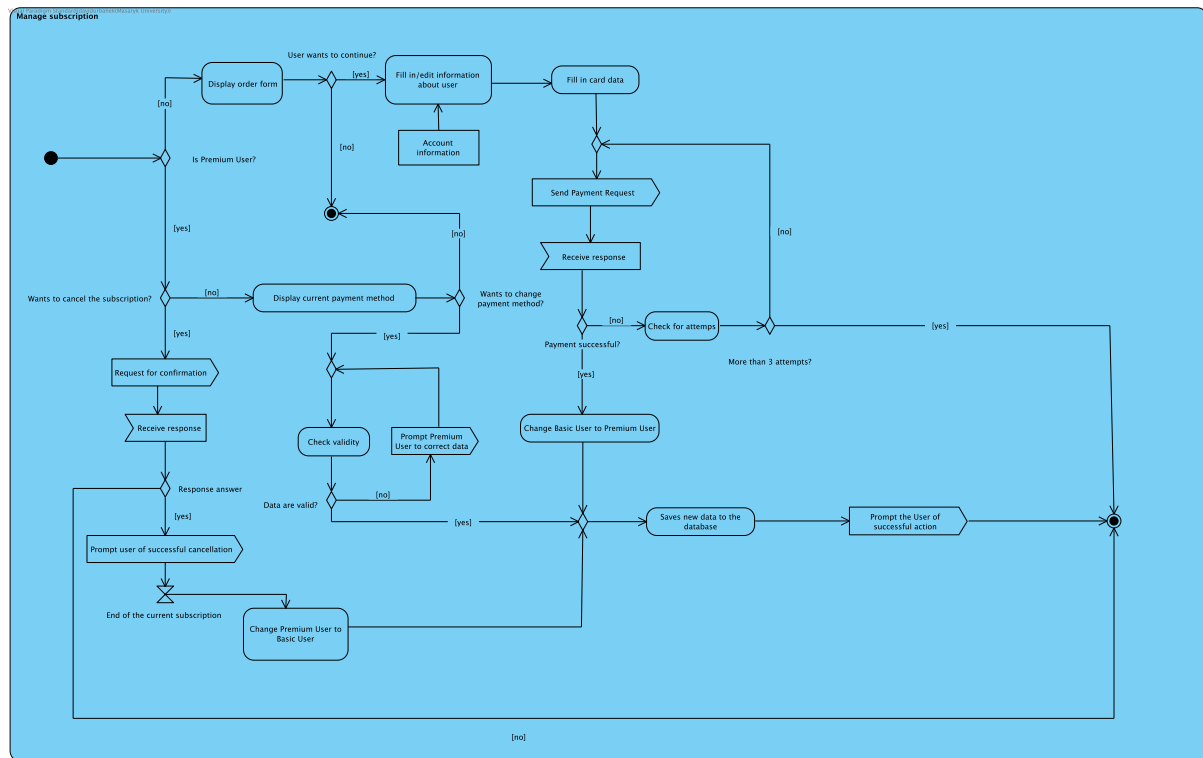
Details

Regions

Name	Description
 <code>setId(id) / id = id</code>	

Activity Diagram

manage-subscription



Details

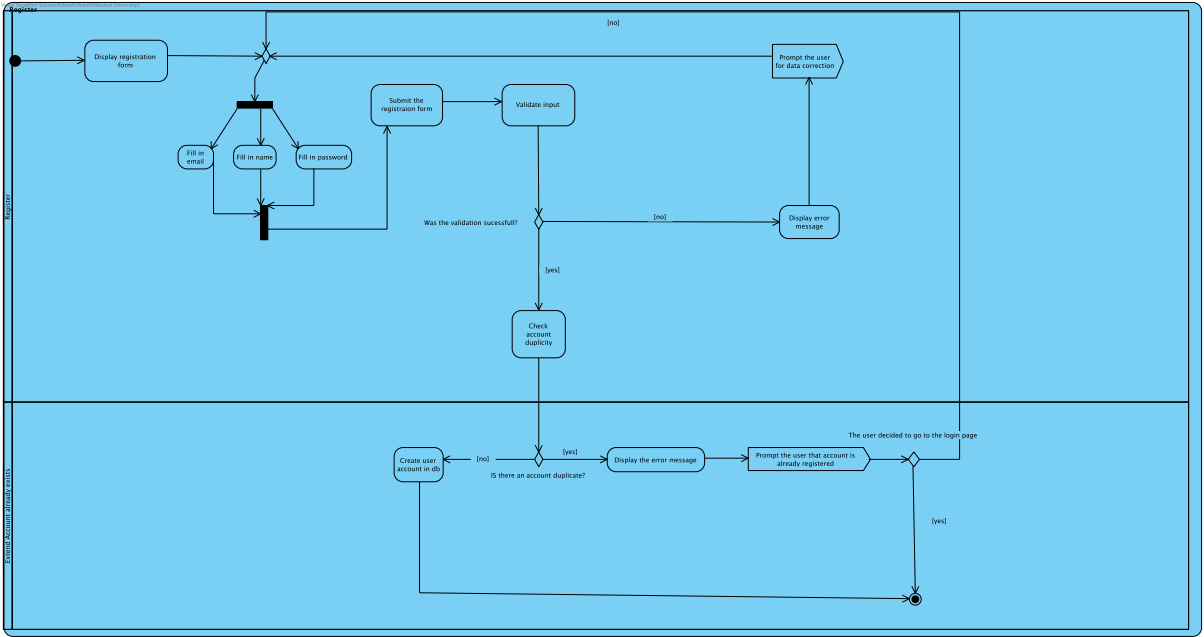
Children

Name	Description
● N/A	
➤ Is Premium User?	
➤ Fill in/edit information about user	
➤ Account information	
➤ Fill in card data	
➤ N/A	
➤ Send Payment Request	
➤ Receive response	
➤ Payment successful?	
➤ Change Basic User to Premium User	
● N/A	
➤ More than 3 attempts?	
➤ Wants to cancel the subscription?	
➤ Request for confirmation	
➤ Receive response	
➤ Response answer	

 Prompt user of successful cancellation	
 End of the current subscription	
 Change Premium User to Basic User	
 Display current payment method	
 Check validity	
 Data are valid?	
 Saves new data to the database	
 N/A	
 Prompt Premium User to correct data	
 Check for attempts	
 Display order form	
 User wants to continue?	
 N/A	
 Wants to change payment method?	
 N/A	
 Prompt the User of successful action	

Activity Diagram

Register



Details

 **Swimlane**

Activity Partitions

Register		
Backlog Activity Id	0	
Contained Elements	Name	Description
	 N/A	
	 Fill in email	
	 Fill in name	
	 Fill in password	
	 N/A	

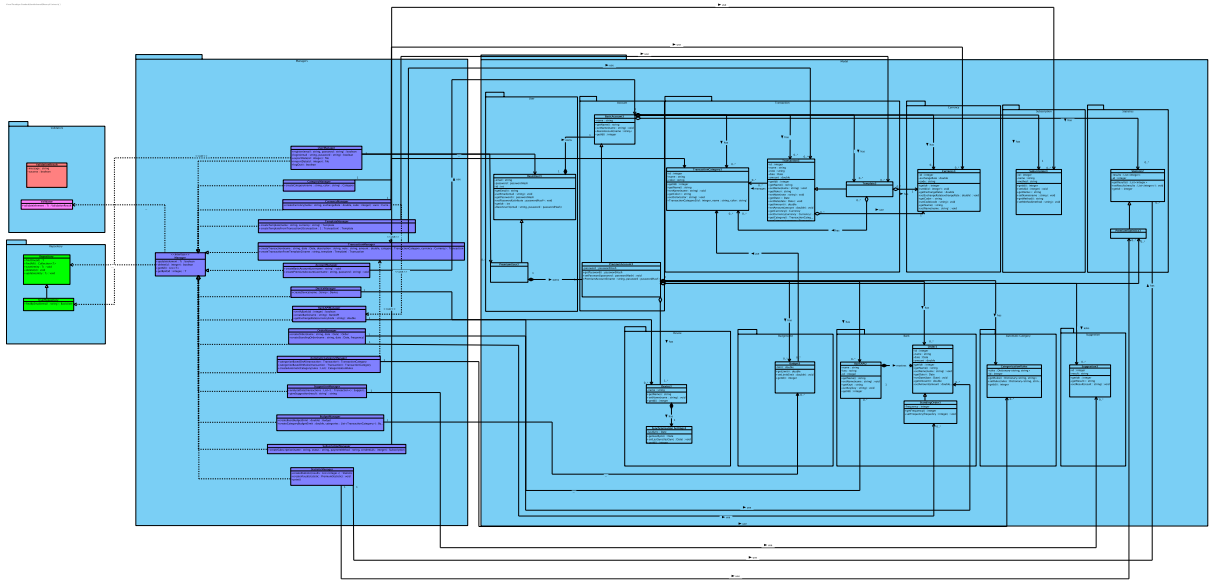
Extend Account already exists		
Backlog Activity Id	0	

Children

Name	Description
 N/A	
 Display registration form	
 Submit the registraion form	
 Validate input	
 Was the validation sucessfull?	
 Display error message	
 Check account duplicity	
 Prompt the user for data correction	
 N/A	
 IS there an account duplicate?	
 Create user account in db	
 N/A	
 Prompt the user that account is already registered	
 The user decided to go to the login page	
 Display the error message	
 Swimlane	

Package Diagram

Package Diagram



Details

Children

Name	Description
 User	
 Account	
 Transaction	
 Currency	
 Subscription	
 Statistics	
 Device	
 Suggestion	
 BudgetLimit	
 Bank	
 Automatic Category	

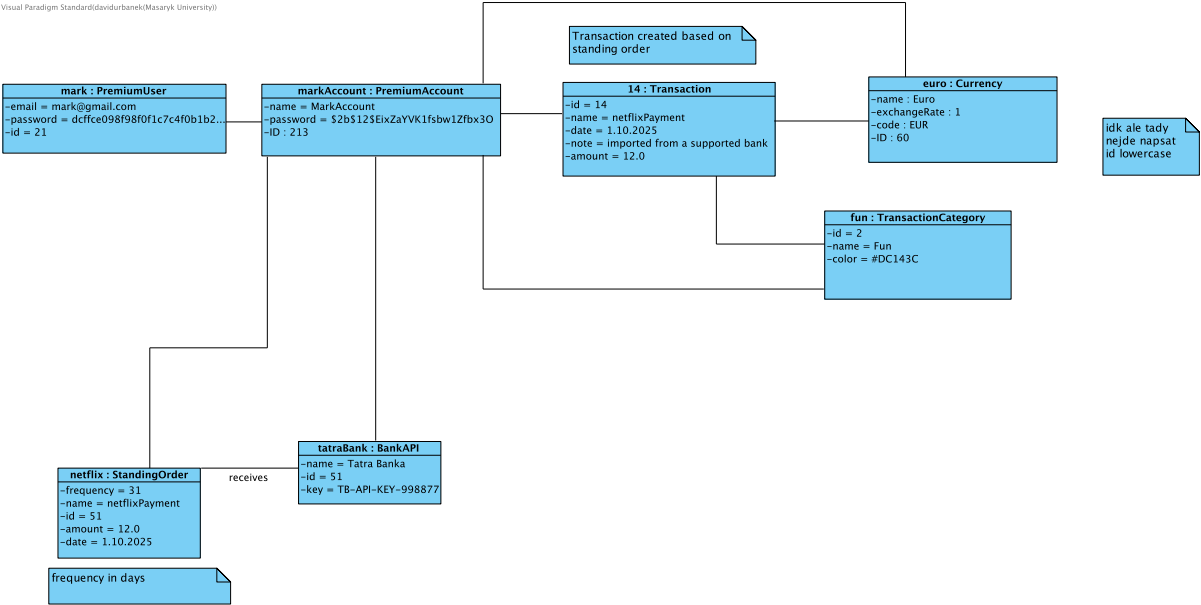
Children

Name	Description
 Package	

Object Diagram

Object Diagram2

Visual Paradigm Standard(davidurbaneK(Masaryk University))

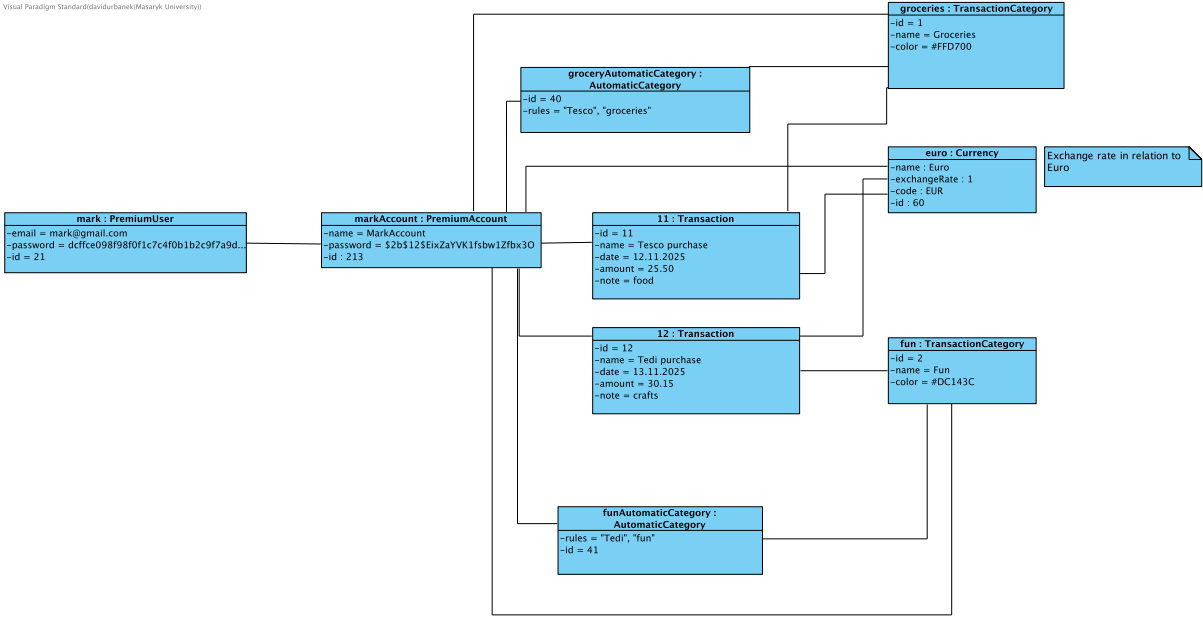


Details

Object Diagram

Transaction

Visual Paradigm Standard (davidurbanek@macsaryk.university.pl)



Details

Entity Relationship Diagram1

